Effects of Underflow on Solving Linear Systems (Excerpt)

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Abstract

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This is an excerpt, prepared by Andrew W. Appel in 2025, of some parts of the tech. report relevant to accuracy bounds for forward substitution and back substitution.

In this paper we examine the effects of underflow on solving systems of linear equations using Gaussian Elimination. Our goal is to decide if reliable software for solving linear systems in the presence of underflow can be written at reasonable cost. We constrast the utilities of gradual underflow and "store zero", and show that only by using gradual underflow can we achieve reliability easily.

1. Introduction

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2. Examples

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Let ϵ be the rounding error of the arithmetic, and λ be the underflow threshold. Suppose, for example, a binary floating point number is represented as $f \cdot 2^e$ where f lies between 1 and 2 with an n bit fraction, and e is an integer exponent in the range $e_{\min} \leq e \leq e_{\max}$. Then $\epsilon = 2^{-n}$ (the difference between 1 and the next largest number). The underflow threshold is $\lambda = 2^{e_{\min}}$; this is the smallest . . . normalized number for G.U. . . . Denormalized numbers lie between 0 and $2^{e_{\min}}$ in G.U. arithmetic for which the smallest nonzero number is thus $\mu = \epsilon\lambda$. We will give a more precise model of rounding and underflow errors in the next section.

3. Error Analysis

. . .

Let $\blacksquare$ be one of the operations $(+, -, *, /)$ and let $fl(a\blacksquare b)$ denote the floating point result of the indicated computation. . . . To take underflow into account, we write ([Kahan and Palmer, 1979])

$$fl(a\blacksquare b) = (a\blacksquare b)(1+e) + \eta \qquad \text{unless } a\blacksquare b \text{ overflows.} \tag{11}$$

In the case of G.U. we have the following constraints on e and η:

(1) $|e| \leq \epsilon$ and $|\eta| \leq \lambda\epsilon$,

(2) $\eta \times e = 0$

(3) $\eta = 0$ if $\blacksquare$ is either addition or subtraction.

. . .

We define the function $UN(x)$ to be

$$UN(x) = \begin{cases} 1 & \text{if } x \text{ underflows } (x < \lambda) \\ 0 & \text{if it does not} \end{cases}$$

We assume no overflow occurs.

We ignore terms which are $O(\epsilon^2)$ or $O(\epsilon\mu)$. This allows us to make the following substitutions when convenient:

Replace $1/(1+e_1)$ by $1+e_2$,

Replace $(1+e_1)^j$ by $1+je_2$,

Replace $\prod_{j=1}^{m}(1+e_j)/\prod_{i=1}^{n}(1+e_i')$ by $1+(n+m)e$

Replace $\eta_1 * (1+e_1)^j$ by η_2.

By replacing every appearance of an original datum a by $1.\cdot a$, and using the above formula for error in multiplication, we can take into account effects of rounding and underflow errors in the original data.[1]

$\|A\|$ and $\|b\|$ denote the norms of matrix A and vector b. $\|A\|_\infty$ denotes the infinity norm of A, and similarly for $\|b\|_\infty$. $k(A) = \|A\|_\infty \|A^{-1}\|_\infty$ denotes the condition number of the matrix A.

$|A|$ ($|b|$) denotes the matrix (vector) whose entries are the absolute values of the corresponding entries in A (b). Inequalities like $|A| > |B|$ ($|a| > |b|$) are meant componentwise.

3.2 Approach

As stated in the introduction, we use backward error analysis. Thus, when trying to solve

$$Ax = b \tag{12}$$

for x we obtain an approximation $\hat{x} = x\ \delta x$ which satisfies the perturbed problem

$$(A + \delta A)\hat{x} = b + \delta b\ . \tag{13}$$

The task of backwards error analysis is to obtain bounds on δA and δb. These bounds can be used in turn to bound the residual r:

$$r = A\hat{x} - b = -\delta A\hat{x} + \delta b \tag{14}$$

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3.3 Error Analysis Details

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Proposition 2.

To analyze solving a triangular system, we need another expression for the error in computing an inner product:

$$\mathit{float}(\sum_{i=1}^{n} a_i b_i) = \sum_{i=1}^{n} a_i b_i (1 + E_i) + \eta\ . \tag{31}$$

In the absence of underflow we have

$$\begin{array}{l} |E_1| \le n\epsilon \\ |E_j| \le (n+2-j)\epsilon \quad \text{if} \quad j > 1\ . \end{array} \tag{32}$$

In the case of G.U.[2] we have the same bounds on the $|E_i|$, and

$$|\eta| \le n\lambda\epsilon \tag{34}$$

[1]Editors note: Ugh. It's provable that $1.\cdot a = a$ exactly, so this notational trick does not appeal to me.

[2]Editor's note: G.U. = Gradual Underflow, that is, a floating-point format that supports subnormal numbers. The technical report also covers the case of "S.Z." or "store zero", which we omit here in all lemmas and theorems.

Proof. Letting

$$\begin{aligned} S_m &= fl(\sum_{i=1}^{m} a_i b_i) \\ &= fl(S_{m-1} + a_m b_m) \\ &= (S_{m-1} + a_m b_m(1+e_m) + \eta_m)(1+w_{m-1}) + z_{m-1} \end{aligned}$$

It is easy to prove by induction on m that

$$\begin{aligned} S_m = \ & a_1 b_1 (1+e_1) \prod_{i=1}^{m-1} (1+w_i) + \eta_1 \prod_{i=1}^{m-1} (1+w_i) + \\ & \sum_{k=2}^{m} a_k b_k (1+e_k) \prod_{i=k-1}^{m-1} (1+w_i) + \sum_{k=2}^{m} \eta_k \prod_{i=k-1}^{m-1} (1+w_i) + \\ & \sum_{k=1}^{m-1} z_k \prod_{i=k+1}^{m-1} (1+w_i). \end{aligned}$$

$z_{m-1} = 0$ always for G.U. ... Thus ... $z_{m-1}\eta_m = 0$, so the contribution from underflow is at most n times the bound for η_m and z_{m-1}. Approximating $(1+e_k)\prod_{i=r}^{n-1}(1+w_i)$ by $1+(n-r+1)\epsilon$ we get the result. Note that the $\sum_{i=1}^{n} a_i b_i (1+E_i)$ term is the same term as would have been given by round-off alone. Q.E.D.

...

Lemma 4: Error Analysis of Solving a Triangular System

Consider the lower triangular system $Ly = b$ where L does not necessarily have ones on the diagonal. (The analyses when only ones are on the diagonal or L is upper triangular are similar.) Then in solving $Ly = b$ using the program in Appendix 2 (where inner products are accumulated with roundoff error ϵ_s and smallest number μ_s we obtain $\hat{y} = y + \delta y$ which actually satisfies $(L + \delta L)\hat{y} = b + \delta b$, where

$$|\delta L_{ij}| \le |L_{ij}| \cdot \begin{cases} \epsilon & \text{if} \quad i = j = 1 \\ (i-1)\epsilon_s & \text{if} \quad i > j = 1 \\ (i+1-j)\epsilon_s & \text{if} \quad i > j > 1 \\ \epsilon + \epsilon_s & \text{if} \quad i = j > 1 \end{cases} \tag{80}$$

independent of underflow.

In the absence of underflow

$$\delta b = 0 \ . \tag{81}$$

Using G.U.

$$\begin{array}{llll} \text{if} \quad i = 1 & |\delta b_i| & \le \lambda\epsilon |L_{11}| UN(y_1) & \text{if } y_1 \text{ underflows} \\ \text{if} \quad i > 1 & |\delta b_i| & \le i \cdot \lambda_s \epsilon_s + & \text{intermediate underflows} \\ & & \lambda\epsilon |L_{ii}| UN(y_i) & \text{if } y_i \text{ underflows} \end{array} \ . \tag{82}$$

If L_{ii} is known to be 1, the only changes in the above bounds (besides substituting 1 for L_{ii}) are that $E_{11} = \delta b_1 = 0$ independent of underflow.

Proof: We apply Proposition 2 to the program in Appendix 2. We have two cases.
Case $i = 1$: In this case,

$$y_1 = float(b_1/L_{11}) = \frac{b_1}{L_{11}(1+E_{11})} + \nu_1 \tag{84}$$

so

$$L_{11}(1+E_{11})y_1 = b_1 + \nu_1 L_{11}(1+E_{11}) UN(y_1) \tag{85}$$

where

$$|E_{11}| \le \epsilon \quad \text{and} \quad |\nu_1| \le \lambda \text{ or } \lambda/\epsilon \tag{86}$$

Case $i > 1$: Now

$$\begin{aligned} y_i &= \mathit{float}((b_i - \textstyle\sum_{j=1}^{i-1} L_{ij} y_j)/L_{ii}) \\ &= [(1+e_i)(b_i - \textstyle\sum_{j=1}^{i-1} L_{ij}(1+E_{ij})y_j - \rho_i) + \sigma_i]/(L_{ii}(1+\zeta_i)) + \nu_i \end{aligned} \tag{87}$$

where

$$|E_{ij}| \le \begin{cases} (i-1)\epsilon_{\rm s} & \text{if } \; j = 1 \\ (i+1-j)\epsilon_{\rm s} & \text{if } \; j > 1 \end{cases} \tag{88a}$$

$$|e_i| \le \epsilon_{\rm s} \tag{88b}$$

$$|\zeta_i| \le \epsilon \tag{88c}$$

and

$$|\rho_i| \le (i-1)\lambda_{\rm s}\epsilon_{\rm s} \qquad \text{using G.U.} \tag{89a}$$

$$|\sigma_i| \le \lambda_{\rm s}\epsilon_{\rm s} \qquad \text{using G.U.} \tag{89b}$$

$$|\nu_i| \le \lambda\epsilon \qquad \text{using G.U.} \tag{89c}$$

In the absence of underflow, $\rho_i = \sigma_i = \nu_i = 0$. Rearranging equation (87), we obtain

$$L_{ii}(1+\zeta_i)y_i = (1+e_i)(b_i - \sum_{j=1}^{i-1} L_{ij}(1+E_{ij})y_j - \rho_i) + \sigma_i + \nu_i L_{ii}(1+\zeta_i) \tag{90}$$

or

$$b_i = \sum_{j=1}^{i-1} L_{ij}(1+E_{ij})y_j + L_{ii}\left(\frac{1+\zeta_i}{1+e_i}\right) y_i + \rho_i - \frac{\sigma_i}{1+e_i} - \nu_i L_{ii}\left(\frac{1+\zeta_i}{1+e_i}\right) \quad . \tag{91}$$

Using the bounds in (88) and (89) in (91), we obtain the bounds in the statement of the lemma. Q.E.D.